

HOLDING VIRTUAL REALITY BACK

The main hurdles in the mass adoption of Virtual Reality technologies



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With Metaverse being the latest buzzword everyone is talking about, the focus on VR has also risen significantly. VR will likely be the early catalyst in allowing for the vision of the metaverse to take off since it allows individuals to experience virtual environments in a truly immersive manner. Hence, virtual reality should present our senses with a version of reality that isn't really there, but from our view, is as close to reality as can be. However, since its inception, virtual reality has been faced with deterrents on the road to mass adoption. Before we delve into the nitty-gritty of the challenges Virtual Reality faces in today's world, it is essential to explain the concept first, especially for the uninitiated.

Virtual Reality is the term used to describe a three-dimensional, virtual environment (mostly computer-

generated) that can be interacted with or explored by users. In the ideal situation, users should even be able to manipulate objects or perform actions within the space. Numerous industries including medicine, gaming, manufacturing, and others have tried incorporating VR technology within their processes with various degrees of success. The major hurdle has always been mass adoption though. Films such as Ready Player One have already given people a taste of what VR should look and feel like, but we're honestly a ways away from that. And why is that even though VR was being worked on way back in the 1980s? This is where we come back to the challenges virtual reality faces. So, let's get right into the hurdles VR faces.

HARDWARE ISSUES AND COSTS

According to the HTC Vive 2020 Developer Survey, four in ten developers shared that their biggest concern relating to VR hardware was the cost of the hardware to end-users.

It is no secret that one of VR's biggest issues is the price tag attached to it. A brand new Oculus Quest 2 will set you back by ₹40,000, which is a pretty decent price when you look at it, but it is still high enough that people will be apprehensive to dive in. Now, while the Quest 2 is designed as a standalone VR headset that doesn't need to be hooked up to a PC or laptop to play games, hooking one up can open up an entire library of PC-only content. Also, games played through Oculus Link on a high-end PC are mostly superior to the native Quest 2 versions when it comes to graphics.

Also, other popular VR headsets such as the Oculus Rift and HTC Vive do need to be connected to a PC, which increases the cost of entry by a significant amount. When we say significant, we mean it, since you would need to own a PC or laptop with high-enough specs in order to play the graphically-intensive VR games. Another issue with the headsets that run by being hooked up to PCs is the cords. While these headsets have some great capabilities in terms of hardware, the cords and excessive assistive devices can quickly clutter up a room and reduce mobility, and in turn, immersiveness. In games where people have to move around, this can prove to be a hassle



HTC VIVE Pro HMD setup

and even, a safety hazard! Nevertheless, the introduction of standalone headsets such as the Oculus Quest 2 at a pretty decent price of \$299 shows that the VR industry is headed in the right direction. We can only hope for standalone headsets that will be able to run more graphically-intensive games in the future.